SIM2001 Advanced Calculus : Tutorial 3

1. Find all the local maxima, local minima and saddle points of the functions.

   (13.7.9) $f(x,y) = x^2 - y^2 - 2x + 4y + 6$

   (13.7.17) $f(x,y) = x^3 + 3xy^2 - 15x + y^3 - 15y$

   (13.7.19) $f(x,y) = 4xy - x^4 - y^4$

   (13.7.21) $f(x,y) = \dfrac{1}{x^2 + y^2 - 1}$

   (13.7.23) $f(x,y) = y \sin x$

   (13.7.27) $f(x,y) = e^{-y}(x^2 + y^2)$

   (13.7.29) $f(x,y) = 2\ln x + \ln y - 4x - y$

2. Find the absolute maxima and minima of the functions on the given domians.

   (13.7.31) $f(x,y) = 2x^2 - 4x + y^2 - 4y + 1$ on the closed triangular plate bounded by the lines $x = 0$, $y = 2$, $y = 2x$ in the first quadrant

   (13.7.35) $T(x,y) = x^2 + xy + y^2 - 6x + 2$ on the rectangular plate $0 \le x \le 5$, $-3 \le y \le 0$

3. (13.8.1)
   Find the points on the ellipse $x^2 + 2y^2 = 1$ where $f(x,y) = xy$ has its extreme values.

4. (13.8.3)
   Find the maximum value of $f(x,y) = 49 - x^2 - y^2$ on the line $x + 3y = 10$.

5. (13.8.5)
   Find the points on the curve $xy^2 = 54$ nearest the origin.

6. (13.8.11)
   Use the method of Lagrange multipliers to find the dimensions of the rectangle of greatest area that can be inscribed in the ellipse $\dfrac{x^2}{16} + \dfrac{y^2}{9} = 1$ with sides parallel to the coordinates axes.

7. (13.8.19)
   Find the minimum distance from the surface $x^2 - y^2 - z^2 = 1$ to the origin.

8. (13.8.23)
   Find the maximum and minumum values of $f(x,y,z) = x - 2y + 5z$ on the sphere $x^2 + y^2 + z^2 = 30$.

9. (13.8.37)
   Maximize the function $f(x,y,z) = x^2 + 2y - z^2$ subject to the constraints $2x - y = 0$ and $y + z = 0$.

10. (13.8.41)
    Find the extreme values of $f(x,y,z) = x^2yz + 1$ on the intersection of the plane $z = 1$ with the sphere $x^2 + y^2 + z^2 = 10$.